

Aggressive feature reduction degrades the selective reliability and clinical net benefit of clinical risk models

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ABSTRACT

Feature reduction is widely adopted in clinical risk modelling, usually justified by a negligible loss of discrimination (AUROC). Discrimination, however, does not capture what a deployed risk model is relied upon for: clinicians act on calibrated probabilities, a triage model must order its own errors, and equitable care requires reliability across patient subgroups. We introduce a pre-registered, leakage-safe protocol that certifies a reduced model on these properties rather than on discrimination alone, and apply it to twelve public clinical datasets ($n = 155\text{--}6,000$), three feature-ranking methods, and three learners, sweeping the retained-feature fraction from 100 % to 25 % with paired, multiplicity-corrected, bootstrapped comparisons. At a quarter of the features, degradation was material in most of the 108 dataset–ranker–learner cells: selective reliability worse in 93, clinical net benefit in 78, conformal set size larger in 80, and discrimination in 90, whereas calibration error and the subgroup gaps were mixed. The harm was dose-dependent and dataset-dependent, and post-hoc recalibration (Platt or isotonic) repaired only the random forest’s calibration error, recovering none of the selective-reliability or net-benefit loss. Discrimination is therefore an insufficient safety certificate for a reduced model; calibration, selective reliability, clinical utility, and subgroup equity should be audited per dataset and learner.

1 Introduction

Predictive models built on tabular clinical data—heart-disease risk, diabetes onset, liver-disease screening, post-heart-failure mortality—are typically reported with a single headline of discrimination, most often the area under the receiver-operating characteristic curve (AUROC). When such models are deployed as decision support, however, discrimination is not the primary property a clinician relies on. A clinician reads a *probability*—“this patient has an 18 % risk”—and the value of that number depends on whether it is *calibrated*: whether, among patients assigned an 18 % risk, close to 18 % actually experience the event. A model can rank patients well (high AUROC) and yet be badly calibrated, systematically over- or under-stating absolute risk, which is precisely the failure mode that misleads treatment thresholds.

Feature reduction is an almost universal preprocessing step. Reducing the feature budget lowers data-collection cost and simplifies deployment, and is typically defended by showing that AUROC changes little when the model is restricted to the top-ranked features. This accuracy-centred justification, however, does not examine two properties of direct clinical importance: calibration and *subgroup reliability*—whether calibration and discrimination hold equally across protected groups such as sex or age. These properties are seldom re-examined after features are dropped. If aggressive reduction degrades calibration or widens a subgroup gap while leaving AUROC almost intact, a model selected on the conventional evidence is less trustworthy and less equitable than its developers would infer from that evidence.

This paper addresses that risk with a methodological contribution: not a new feature-selection method, but a reusable way to *certify* a reduced model. The prevailing evaluation practice certifies feature reduction on discrimination (AUROC) alone; we replace it with a pre-registered, leakage-safe protocol that evaluates a reduced model on the properties a deployed risk score is actually trusted for—calibration, selective (risk–coverage) reliability, and between-subgroup equity. The protocol quantifies, as the fraction of retained features decreases, the change in calibration (expected calibration error, ECE, and the Brier score), in selective reliability (the area under the risk–coverage curve, AURC, which captures whether the model’s own confidence can be trusted to triage which predictions to act on), in discrimination (AUROC), and in the between-subgroup gaps in AUROC and calibration. We fix the outcomes, the reduction levels, and the decision rule before examining any result, apply the protocol under a strict leakage firewall to twelve public datasets with three base learners and three ranking methods, and report results in both directions: the settings in which reduction is safe and those in which it is not.

Contributions. (i) A reusable, pre-registered, leakage-safe evaluation protocol that certifies a feature-reduced clinical risk model on calibration, selective reliability, and subgroup equity rather than on discrimination alone—the current state-of-the-art justification for feature reduction—with every number regenerable from committed code and recorded seeds. (ii) An instantiation across twelve public clinical datasets, three learners, and three rankers, showing the protocol reveals deployment-relevant degradation that discrimination-only evaluation misses, and that the harm is strongly conditional on dataset and learner—with an exploratory cross-dataset meta-analysis that turns this conditionality into an *a priori* screen for when reduction is safe. (iii) A leakage-safe recalibration control, built into the protocol, that separates *repairable* (calibration-error) from *irreparable* (selective-reliability, Brier) harm, yielding an actionable, both-directions decision rule for practitioners.

2 Related work

Feature selection in clinical machine learning. Filter, embedded, and wrapper feature-selection methods are standard in clinical prediction, and are conventionally evaluated by discrimination or classification accuracy at a reduced feature count^{1,2}. We use three widely adopted rankers as representative of this practice—a filter (mutual information), an embedded tree-importance ranker (random-forest Gini importance³), and an embedded sparse linear ranker (L_1 -penalised logistic regression)—and treat the *ranking* as given, auditing what its aggressive application costs.

Calibration of risk models. Calibration—the agreement between predicted probabilities and observed frequencies—is a long-standing requirement for clinical risk scores, quantified by the Brier score⁴ and by binned expected calibration error^{5,6}. It is well established that good discrimination does not imply good calibration⁷, and that prediction models should be judged by a framework of traditional and novel measures spanning discrimination, calibration, and clinical utility⁸. Our contribution is to treat calibration as a *dependent variable of preprocessing*: we measure how it moves as features are removed, rather than measuring it once for a fixed model.

Selective prediction and the risk–coverage trade-off. Selective (reject-option) classification lets a model abstain on inputs it is least sure about⁹; its quality is summarised by the risk–coverage curve and its area (AURC), which is low when the model’s confidence ranks its errors well¹⁰. We include AURC because it captures a deployment-relevant property—can the model be trusted to triage its own predictions?—that neither AUROC nor a single calibration number expresses.

Algorithmic fairness in health. Subgroup performance gaps—differences in discrimination or calibration across protected groups such as sex or age—are a central concern for equitable clinical AI^{11,12}. Most audits fix the model and compare groups; we instead ask whether a routine engineering choice (feature reduction) *widens* such gaps, connecting the fairness and preprocessing literatures.

Gap. To our knowledge these threads are not usually combined: feature reduction is justified on accuracy, while calibration, selective reliability, and subgroup gaps are audited—if at all—on the final model only. We close that gap with a single pre-registered protocol that tracks all four as a function of the feature budget.

3 Methods

We formalise the audit as a reusable certification protocol whose architecture is summarised in Fig. 1 and specified in Algorithms 1 and 2. Sections 3.1–3.5 define each component; Section 4 instantiates the protocol on twelve datasets, three learners, and three rankers. The protocol takes a clinical dataset and a feature-reduction configuration and returns, for every retained-feature budget, a direction-resolved verdict on each reliability axis, computed under a strict leakage firewall so that no information from a held-out patient can enter preprocessing. The protocol is designed to operationalise the calibration, clinical-utility, and fairness reporting items of TRIPOD+AI, the reporting standard for clinical prediction models using machine learning¹³: it reports calibration (ECE, Brier) and clinical utility (decision-curve net benefit) alongside discrimination, examines performance across protected subgroups, and releases code and a pre-registration so that every reported quantity is reproducible. A completed adherence note is provided in the repository.

3.1 Datasets

We use twelve public clinical tabular datasets (Table 1), spanning cardiology (Cleveland¹⁴, Statlog and SPECTF heart), oncology (the Wisconsin diagnostic¹⁵ and original breast-cancer datasets, mammographic mass, and Haberman survival), metabolic and hepatic disease (Pima diabetes¹⁶, the Indian Liver Patient Dataset, hepatitis), post-heart-failure mortality¹⁷, and a large real-world electronic-health-record cohort, the Diabetes 130-US hospitals dataset (early readmission). Each carries a binary outcome and, where available, a protected attribute that is *never* used as a model feature. The datasets span two orders of magnitude in sample size ($n = 155\text{--}6,000$), feature counts from $p = 3$ to 44, outcome prevalences from 0.11 to 0.79, and protected attributes covering sex, gender, age, and (in the readmission cohort) race, so the audit is not tied to one clinical context and supports a cross-dataset meta-analysis (Section 4.7).

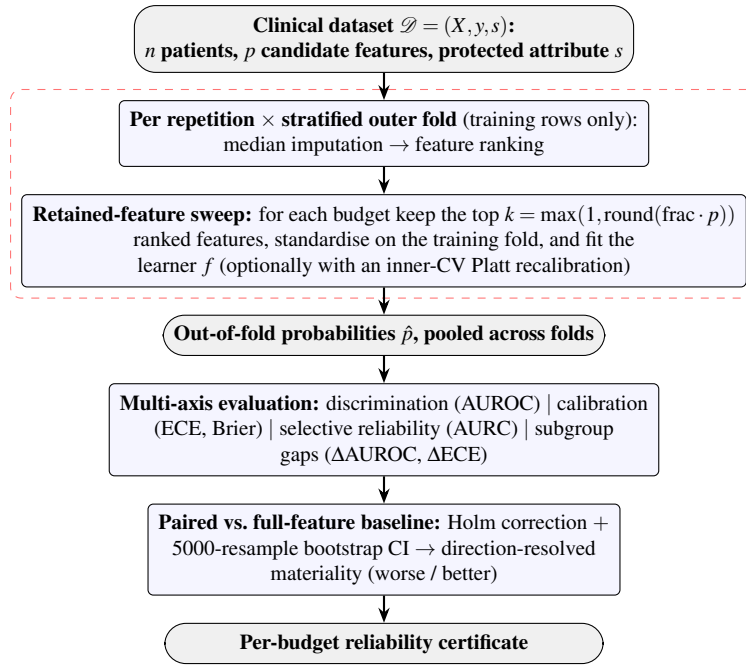


Figure 1. Architecture of the certification protocol. Imputation, feature ranking, standardisation, and model fitting all occur inside the training portion of each cross-validation fold (the dashed *leakage firewall*); only out-of-fold probabilities are scored. Each reliability axis is compared, paired across repetitions, to the full-feature baseline and resolved into a materially-worse or materially-better verdict per budget.

Clinical stakes. Each outcome is acted on at a probability threshold, which is precisely why calibration and selective reliability—not ranking alone—matter clinically. On Cleveland, coronary-disease risk informs statin initiation and further cardiac testing, so a reduced model that over- or under-states absolute risk shifts which patients are treated; sex is the equity axis. Pima diabetes-onset risk drives preventive-intervention and screening decisions, where a deflated risk misses preventable onset; here the calibration–fairness gap is examined across an age stratum. ILPD liver-disease risk informs specialist referral, with gender as the equity axis. Heart-Failure mortality risk informs intensity-of-care and palliative-care discussions—a setting in which a model used to triage must properly order its own errors (AURC). WDBC malignancy probability governs whether a borderline case is flagged for human review; it carries no protected attribute and serves as the wide-feature, reduction-benign anchor. The Diabetes 130-US readmission cohort adds a large electronic-health-record setting in which a 30-day-readmission risk informs discharge and care-transition decisions, with *race* as the equity axis across six groups—the most consequential fairness context in the study. The remaining datasets (Statlog and SPECTF heart, mammographic mass, Haberman survival, hepatitis, original breast cancer) extend the same threshold-decision logic to further diagnostic and prognostic tasks. Across these contexts, a miscalibrated reduced model, or one whose confidence no longer orders its errors, would misallocate the very decisions the score is intended to support.

3.2 Leakage-controlled protocol

For each (dataset, ranking method, base learner, repetition) we run stratified k -fold cross-validation ($k = 5$). Within every training fold, and using training data only, we (i) impute missing values by the training-fold median, (ii) rank all features by the chosen method, and (iii) for each retained fraction keep the top- k features ($k = \max(1, \text{round}(\text{frac} \times p))$), standardise them on the training fold, and fit the base learner. Out-of-fold probabilities are pooled once per (fraction) cell and scored. Because imputation, ranking, scaling, and model fitting all occur inside the training fold, no information from held-out patients can leak into preprocessing. The fraction of retained features is swept over $\{1.0, 0.75, 0.5, 0.33, 0.25\}$, and the same per-repetition cross-validation split is shared across all reduction levels, so every reduced model is compared to its own full-feature baseline as a paired observation. Algorithm 1 states this per-configuration procedure precisely.

3.3 Outcomes

Per cell we compute, all from the pooled out-of-fold probabilities (no extra model fits): **discrimination**—AUROC; **calibration**—15-bin expected calibration error (ECE) and the Brier score (lower is better for both); **selective reliability**—AURC, the area

Table 1. Datasets. Prevalence is the positive-class rate; p is the number of model features (the protected attribute is excluded from the features). All are publicly available; see Data availability.

Dataset	Outcome	n	p	Prev.	Protected
bcw	malignant	699	9	0.345	none
cleveland	coronary disease	303	12	0.459	sex
diabetes130	30-day readmission	6000	16	0.112	race
haberman	5-year death	306	3	0.265	age group
heartfailure	death	299	11	0.321	sex
hepatitis	death	155	18	0.206	sex
ilpd	liver disease	583	9	0.714	gender
mammographic	malignant	961	5	0.463	age group
pima	diabetes onset	768	8	0.349	age group
spectf	abnormal SPECT	267	44	0.794	none
statlogheart	coronary disease	270	12	0.444	sex
wdbc	malignant	569	30	0.373	none

under the risk–coverage curve with confidence taken as $|p - 0.5|$ (lower is better); **clinical utility**—the standardised decision-curve net benefit $NB(t) = \frac{TP}{n} - \frac{FP}{n} \frac{t}{1-t}$ averaged over the threshold range $t \in [0.05, 0.5]$ (higher is better)¹⁸; **distribution-free reliability**—split-conformal prediction sets at 90 % nominal coverage, summarised by the mean set size (efficiency; lower is better); marginal coverage is approximately nominal (exactly so under exchangeability of the pooled out-of-fold scores), and because we compare the reduced and full-feature models built by the identical construction, our reported quantity is the *paired* set-size difference, in which any departure from exchangeability is shared and largely cancels^{19,20}; **selective calibration**—ECE on the 80 % most-confident subset; and **subgroup gaps**—the between-group range (max minus min) of AUROC, ECE, net benefit, and conformal coverage across the protected attribute’s groups (computed only where a group has at least ten samples and both classes present; undefined, and reported as such, for the three datasets without a protected attribute). The primary estimand is the *paired difference* of each outcome at a reduced fraction versus the full-feature baseline; a positive ECE/Brier/AURC/set-size/gap difference, or a negative AUROC/net-benefit difference, means reduction *degrades* that property.

3.4 Statistics

For each (dataset, method, learner, outcome) family across the reduced fractions, paired differences versus the full-feature baseline are tested with a two-sided Wilcoxon signed-rank test and corrected for multiple comparisons by the Holm step-down procedure²¹ (R3). We use the non-parametric Wilcoxon test, rather than a paired t -test, because the per-cell paired differences across heterogeneous datasets are not assumed to be normally distributed and the test is robust to outliers; the alpha level is 0.05 throughout. Each difference carries a 5000-resample percentile bootstrap confidence interval (R12). A difference is called *material* only if its Holm-corrected $p < 0.05$ and its bootstrap CI excludes zero, and is then resolved by the sign of the paired difference into *materially worse* or *materially better* (Algorithm 2). Holm correction thus controls the family-wise error rate *within* each (dataset, method, learner, outcome) family across the four reduced budgets; the aggregate worse/better cell counts we report across the 108 cells are descriptive summaries of these per-family verdicts, not a single study-wide family-wise claim, and we read them as a consistent pattern of degradation rather than as one global hypothesis test. Headline numbers use 30 repetitions per cell (R4).

3.5 Pre-registration and reproducibility

Outcomes, reduction levels, and the materiality rule were frozen before any result was read, in a deposited pre-registration. The study follows a fixed set of reproducibility rules: no number appears unless it regenerates end-to-end from committed code, the calibration file, and a recorded seed (R1); all tables derive from a single canonical aggregator (R2); each cell uses a deterministic per-cell seed (R5); the SHA-256 of the calibration file is embedded in every results CSV and downstream analysis refuses to mix mismatched hashes (R6); raw per-trial outputs are released in summary form with validated/provisional status tags (N14).

4 Results

All numbers below are the paired difference of an outcome at a reduced feature budget versus the full-feature model, averaged over the twelve datasets, three rankers, and three learners unless stated otherwise; a cell is one (dataset, ranker, learner) combination—108 in total—and is *material* when its Holm-corrected $p < 0.05$ and its 5000-bootstrap CI excludes zero.

Algorithm 1 CERTIFYFEATUREREDUCTION: leakage-safe per-configuration evaluation.

Require: dataset (X, y, s) with p features; ranker m ; learner f ; budgets $\mathcal{F}$; repetitions R ; folds K ; master seed σ ; recalibration mode $c \in \{\text{NONE}, \text{PLATT}\}$

Ensure: metric tensor $M[\text{frac}, \text{axis}, r]$ over budgets, reliability axes, repetitions

```

1: for  $r \leftarrow 1$  to  $R$  do
2:    $\rho \leftarrow \text{DERIVESEED}(\sigma, m, f, r)$  ▷ seed shared across budgets  $\Rightarrow$  paired
3:    $\{(\text{tr}_\kappa, \text{te}_\kappa)\}_{\kappa=1}^K \leftarrow \text{STRATIFIEDKFOLD}(X, y, K, \rho)$ 
4:   for each fold  $\kappa$  do ▷ training rows only: the leakage firewall
5:      $\mu_\kappa \leftarrow \text{median}(X_{\text{tr}_\kappa})$ ;  $\tilde{X} \leftarrow \text{IMPUTE}(X, \mu_\kappa)$ ;  $\pi_\kappa \leftarrow \text{RANK}_m(\tilde{X}_{\text{tr}_\kappa}, y_{\text{tr}_\kappa})$ 
6:   end for
7:   for each budget  $\text{frac} \in \mathcal{F}$  do
8:      $k \leftarrow \max(1, \text{round}(\text{frac} \cdot p))$ ;  $\hat{p} \leftarrow$  undefined vector of length  $n$ 
9:     for each fold  $\kappa$  do
10:       $S \leftarrow \pi_\kappa[1:k]$  ▷ top- $k$  ranked features
11:       $g \leftarrow \text{FIT}(f, c, \text{STD}(\tilde{X}_{\text{tr}_\kappa, S}), y_{\text{tr}_\kappa})$  ▷  $c=\text{PLATT}$ : inner-CV recalibration
12:       $\hat{p}_{\text{te}_\kappa} \leftarrow g(\text{STD}(\tilde{X}_{\text{te}_\kappa, S}))$  ▷ standardisation fitted on  $\text{tr}_\kappa$ 
13:    end for
14:     $M[\text{frac}, \cdot, r] \leftarrow (\text{AUROC}, \text{ECE}, \text{BRIER}, \text{AURC}, \Delta_{\text{AUROC}}^s, \Delta_{\text{ECE}}^s)(y, \hat{p}, s)$ 
15:  end for
16: end for
17: return  $M$ 

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Table 2. Paired difference at 25 % of features versus the full-feature model (mean over the twelve datasets, three rankers, and three learners; positive ECE/Brier/AURC/gap values indicate degradation, positive AUROC/net-benefit values indicate improvement). The two right columns split the Holm-significant cells (bootstrap CI excluding zero) by the direction of the difference into materially *worse* and materially *better*; for the subgroup gaps, a widening counts as worse. Computed from the released per-configuration result summary.

Outcome	Mean diff. vs. full	Material worse	Material better
AUROC ($\uparrow$ better)	-0.0390	90/108	4/108
ECE ($\downarrow$ better)	+0.0019	42/108	39/108
Brier ($\downarrow$ better)	+0.0124	77/108	20/108
AURC ($\downarrow$ better)	+0.0316	93/108	8/108
AUROC subgroup gap ($\downarrow$ better)	+0.0017	16/72 (widened)	21/72
ECE subgroup gap ($\downarrow$ better)	-0.0019	19/72 (widened)	23/72

4.1 Degradation is dose-dependent

Table 2 reports the most aggressive budget (25 % of features). At that level, selective reliability, discrimination, clinical net benefit, and conformal informativeness are materially worse than the full-feature model in the large majority of the 108 cells (AURC in 93, AUROC in 90, decision-curve net benefit in 78, conformal set size larger in 80, and the Brier score in 77), while probability calibration is mixed (ECE materially worse in 42, better in 39). The effect grows monotonically as features are removed (Fig. 2): the mean AURC penalty rises from +0.002 at 75 % of features (materially worse in 50 of 108 cells) to +0.032 at 25 % (93 of 108), and the AUROC drop from -0.002 to -0.039 (45 then 90 of 108), with net benefit tracking the same trajectory (-0.001 to -0.012).

4.2 Calibration and selective reliability do not merely track discrimination

A natural hypothesis is that the selective-reliability and net-benefit loss is attributable to the discrimination loss alone. Three observations argue against this. First, the reliability loss matches or exceeds the discrimination loss in extent: AURC is materially worse in 93 of 108 cells and net benefit in 78, against 90 for AUROC, with the Brier score comparable (77); the degradation thus extends beyond ranking. Second, the axes decouple: at every budget there are cells in which calibration or selective reliability is materially worse while AUROC is not (16, 11, 12, and 6 of 108 at the 75, 50, 33, and 25 % budgets; Table 3), largest at the mildest budget—the regime in which reduction is applied most readily. Third, probability calibration error behaves differently from the rest: across the twelve datasets ECE is mixed (materially worse in 42 cells but better in 39 at the most aggressive budget), whereas selective reliability, net benefit, and the Brier score degrade consistently. On several

Algorithm 2 AGGREGATEANDDECIDE: direction-resolved materiality.

Require: metrics M (Alg. 1) for all configurations; reference budget $\text{frac}_0=1.0$; level $\alpha=0.05$; resamples $B=5000$; per-axis degradation sign $\delta_o \in \{+1, -1\}$

Ensure: verdict $\in \{\text{WORSE}, \text{BETTER}, \text{N.S.}\}$ per (configuration, budget, axis)

```
1: for each configuration and axis  $o$  do
2:    $P \leftarrow \langle \rangle$  ▷ ordered raw  $p$ -values for the Holm family
3:   for each budget  $\text{frac} \neq \text{frac}_0$  do
4:      $d \leftarrow M[\text{frac}, o, \cdot] - M[\text{frac}_0, o, \cdot]$  ▷ paired across repetitions
5:      $p_{\text{raw}} \leftarrow \text{WILCOXON}(d)$ ;  $[\ell, u] \leftarrow \text{BOOTSTRAPCI}(d, B)$ ; append  $p_{\text{raw}}$  to  $P$ 
6:   end for
7:    $\{p_{\text{holm}}\} \leftarrow \text{HOLMSTEPDOWN}(P)$ 
8:   for each budget  $\text{frac} \neq \text{frac}_0$  do
9:      $\text{material} \leftarrow (p_{\text{holm}} < \alpha) \wedge (0 \notin [\ell, u])$ 
10:    if  $\text{material} \wedge (\text{sign}(\bar{d}) = \delta_o)$  then verdict  $\leftarrow \text{WORSE}$ 
11:    else if  $\text{material}$  then verdict  $\leftarrow \text{BETTER}$ 
12:    else verdict  $\leftarrow \text{N.S.}$ 
13:    end if
14:  end for
15: end for
```

Table 3. Cells (dataset, ranker, learner) in which calibration or selective reliability (ECE, Brier, or AUROC) degrades materially while AUROC does not, by retained-feature fraction. These dissociations are infrequent and concentrated at mild reduction levels, where AUROC itself rarely changes materially. Computed from the released per-configuration result summary.

Fraction of features retained	AUROC-silent cells (of 108)
0.75	16
0.5	11
0.33	12
0.25	6

datasets aggressive reduction even *improves* the calibration error while degrading the model’s selective reliability and clinical net benefit—so calibration error and the deployment-relevant reliability of the model’s own confidence are distinct properties, and an audit that checks only one can miss the other.

4.3 The harm varies substantially across datasets

The harm is far from uniform (Fig. 2, Table 4). At 25 % of features the AUROC penalty spans more than a thirty-fold range, from +0.070 on mammographic mass and +0.068 on Statlog heart down to +0.002 on SPECTF and +0.003 on WDBC. The robust datasets are those with many candidate features (WDBC, 30; SPECTF, 44; the readmission cohort, 16), so a quarter still spans the signal; the harmed datasets have few features (mammographic, 5; Haberman, 3; Statlog and Cleveland, 12), each carrying largely unique signal that reduction cannot afford to discard. The pattern is therefore conditional: aggressive reduction is comparatively benign when the retained budget still spans the predictive signal and harmful when it does not—a relationship the next section makes quantitative and predictive.

4.4 Fragility is learner-dependent

The fragility is not the same across the three learners. At 25 % of features the random forest is the most fragile on every axis (AUROC +0.048, ECE +0.022, net benefit −0.019, AUROC −0.054); logistic regression is the most robust on selective reliability (AUROC +0.019); and gradient boosting is intermediate on reliability (AUROC +0.028) but distinctive on calibration—its ECE actually *improves* under reduction (−0.018). This learner-level split is what makes the pooled ECE result mixed: the boosted model’s already-sharp probabilities tend to tighten as redundant features are dropped, while the forest’s loosen. The random-forest comparison also carries a confound, since forests produce less well-calibrated probabilities out of the box, inflating their apparent ECE penalty. The recalibration control in Section 4.8 disentangles it, showing that most of the forest’s extra calibration-error penalty is an artifact of its uncalibrated probability scale—while the selective-reliability penalty, shared by all three learners, is not.

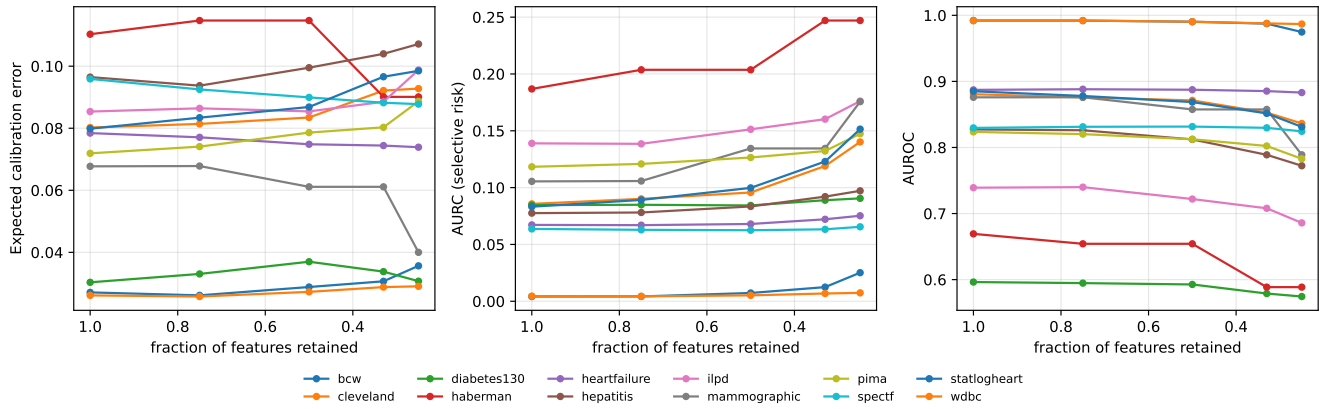


Figure 2. Calibration (ECE), selective reliability (AUC), and discrimination (AUROC) as a function of the fraction of features retained. The horizontal axis is reversed, so more aggressive reduction is to the right; one line per dataset, averaged over rankers and learners. Degradation emerges earliest and most steeply on the low-dimensional datasets (mammographic mass, Haberman, Statlog and Cleveland heart), whereas the wide-feature datasets (WDBC, SPECTF) and the large readmission cohort show minimal change—a pattern quantified by the meta-analysis (Section 4.7).

Table 4. Mean paired difference versus the full-feature model at 25 % of features, per dataset (averaged over rankers and learners; positive ECE/Brier/AUC values indicate degradation, as do negative AUROC values). Computed from the released per-configuration result summary.

Dataset	AUROC	ECE	Brier	AUC
bcw	-0.017	+0.009	+0.016	+0.021
cleveland	-0.044	+0.013	+0.023	+0.055
diabetes130	-0.022	+0.000	+0.002	+0.006
haberman	-0.081	-0.020	+0.002	+0.060
heartfailure	-0.004	-0.005	-0.003	+0.008
hepatitis	-0.055	+0.011	+0.017	+0.020
ilpd	-0.053	+0.013	+0.016	+0.037
mammographic	-0.087	-0.028	+0.021	+0.070
pima	-0.041	+0.017	+0.016	+0.029
spectf	-0.005	-0.008	+0.000	+0.002
statlogheart	-0.054	+0.019	+0.028	+0.068
wdbc	-0.005	+0.003	+0.010	+0.003

4.5 Subgroup gaps: a dataset-specific, not systematic, fairness signal

Across the nine datasets that carry a protected attribute, the subgroup gaps do *not* move in a single direction. At the most aggressive budget the calibration-fairness gap (ECE gap) is essentially a wash on average (-0.002), materially widening in 19 of 72 cells and narrowing in 23; the AUROC subgroup gap is similar ($+0.002$ mean; 16 widen, 21 narrow). The two new fairness axes agree: the net-benefit parity gap²² widens in 18 of 72 group-bearing cells and narrows in 25, and the group-conditional conformal coverage gap widens in 21 of 81 and narrows in 15. The honest reading is that aggressive reduction *can* widen a fairness gap—and does so materially on individual datasets (most consequentially in the race-stratified readmission cohort, where any widening is clinically salient)—but it does not do so systematically across the broader dataset set. Fairness under reduction is therefore a per-dataset audit item, not a property that can be assumed in either direction.

4.6 Reduction reduces clinical utility and conformal informativeness

Two of the new axes translate the abstract reliability loss into clinically actionable quantities, and both degrade (Table 5). Decision-curve net benefit¹⁸—the standardised benefit of acting on the score, averaged over a clinically relevant threshold range—is materially lower at 25 % of features in 78 of 108 cells (mean -0.012), so a reduced model with preserved AUROC nonetheless yields fewer correct treatment or non-treatment decisions per patient. Split-conformal prediction sets^{19,20}, calibrated to 90 % coverage from the out-of-fold scores, become less informative: the mean set size grows in 80 of 108 cells ($+0.070$)—the

Table 5. The four new reliability axes at 25 % of features versus the full-feature model (mean over twelve datasets, three rankers, three learners; the gaps are over the nine group-bearing datasets—the ECE, AUROC, and net-benefit gaps over the eight with a valid two-class subgroup, 72 cells, and the conformal-coverage gap over all nine, 81 cells). Net benefit is a clinical-utility gain ($\uparrow$ better); the others are costs ($\downarrow$ better). Computed from the released per-configuration result summary.

Reliability axis	Mean diff. vs. full	Material worse	Material better
Net benefit (DCA) ($\uparrow$ better)	-0.0125	78/108	15/108
Net-benefit parity gap ($\downarrow$ better)	-0.0021	18/72	25/72
Conformal set size ($\downarrow$ better)	+0.0696	80/108	7/108
Group coverage gap ($\downarrow$ better)	+0.0045	21/81	15/81
Selective ECE ($\downarrow$ better)	+0.0021	45/108	41/108

Table 6. Dataset properties and the reliability harm (AURC penalty at 25 % of features), sorted by harm. The harm is predicted by the feature count (p) and the intrinsic-dim fraction (see text). Computed from the released per-configuration result summary and the dataset definitions.

Dataset	p	$n:p$	Redund.	Intr.dim	Prev.	AURC penalty
mammographic	5	192.20	0.21	1.00	0.46	+0.070
statlogheart	12	22.50	0.17	0.92	0.44	+0.068
haberman	3	102.00	0.05	1.00	0.26	+0.060
cleveland	12	25.25	0.17	0.92	0.46	+0.055
ilpd	9	64.78	0.21	0.67	0.71	+0.037
pima	8	96.00	0.17	0.88	0.35	+0.029
bcw	9	77.67	0.60	0.78	0.34	+0.021
hepatitis	18	8.61	0.18	0.89	0.21	+0.020
heartfailure	11	27.18	0.07	1.00	0.32	+0.008
diabetes130	16	375.00	0.09	0.88	0.11	+0.006
wdbc	30	18.97	0.39	0.33	0.37	+0.003
spectf	44	6.07	0.29	0.57	0.79	+0.002

reduced model must more often return the uninformative two-label set to keep its coverage guarantee. Because marginal coverage is held at the nominal 90 % by construction, set size, not coverage, is the informative axis. Selective calibration—ECE on the 80 % of patients the model is most confident about—is mixed (worse in 45 cells, better in 41), echoing the pooled ECE result: in the high-confidence region on which clinicians most rely, aggressive reduction improves and degrades calibration with approximately equal frequency, even as it consistently reduces net benefit and conformal informativeness.

4.7 Predicting when aggressive reduction is safe: a cross-dataset screen

The harm varies by more than thirty-fold across datasets, and this variation is itself predictable. Taking each dataset’s AURC penalty at 25 % of features as the response and relating it to intrinsic dataset properties (Table 6, Fig. 3), the harm is most strongly predicted by the *feature count*: datasets with more candidate features are reliably safer to reduce (Spearman $\rho = -0.73$, $p = 0.007$). The *intrinsic dimensionality*—the fraction of principal components needed to capture 95 % of the feature variance—is the second predictor ($\rho = +0.64$, $p = 0.024$): harm is greatest when the predictive signal is spread across many components and least when it is concentrated in a few. The sample-to-feature ratio, feature redundancy, and outcome prevalence are not significant predictors. With only twelve datasets this analysis is exploratory and we treat it as hypothesis-generating: of the five candidate predictors, the feature-count association survives a Bonferroni correction for multiplicity ($p = 0.007$, i.e. $< 0.05/5$), whereas the intrinsic-dimension association does not ($p = 0.024$) and is reported as corroborating rather than independent evidence. With this caveat, the practical screening criterion is that aggressive reduction is comparatively safe when a dataset has many features whose signal is concentrated (high p , low intrinsic dimension), so that a quarter of them still spans the predictive subspace, and harmful when it has few features each carrying unique, non-redundant signal. This criterion matches the empirical contrast between the wide WDBC, SPECTF, and readmission datasets (negligible harm) and the compact mammographic, Haberman, and Statlog datasets (largest harm), and it provides an *a priori* indication of whether reduction is appropriate before empirical evaluation; without it, identifying the damaging regime requires measuring the reliability axes this protocol audits.

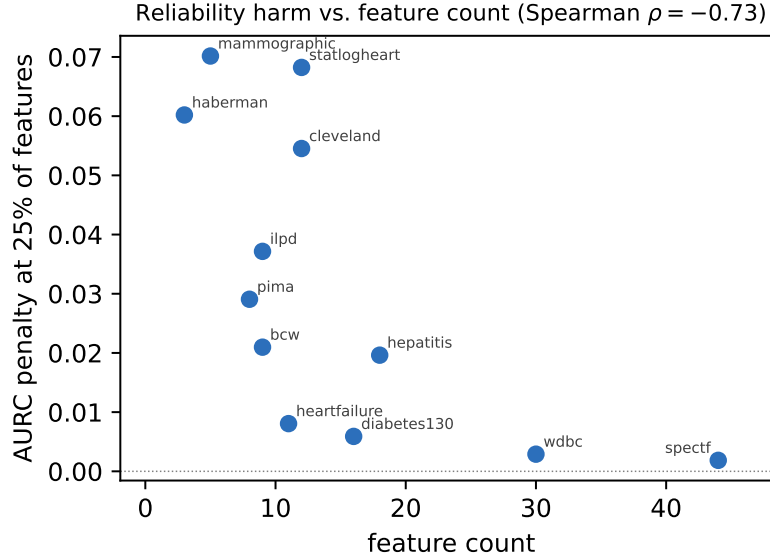


Figure 3. Reliability harm (AUCR penalty at 25 % of features) versus the strongest predictor, the feature count, one point per dataset. Harm falls as the number of candidate features grows: wide datasets tolerate aggressive reduction; compact ones do not.

4.8 Recalibration recovers the calibration error, but not the selective-reliability or net-benefit loss

A standard approach to a calibration problem is post-hoc recalibration, which we evaluate as a control at two depths. In a separate, leakage-safe experiment we wrapped each base learner in either Platt (sigmoid) scaling or the more flexible isotonic regression, each fitted by an inner 3-fold cross-validation *within* each training fold—so the calibrator never sees the outer held-out fold—and recomputed the reduction penalty at the full and 25 % budgets (Table 7). The result is a dissociation with a learner-specific calibration component. Where a raw calibration-error penalty exists, recalibration removes it: the random forest’s ECE penalty collapses from +0.016 to +0.000 (Platt) and +0.002 (isotonic), confirming that the forest’s apparent fragility is largely its uncalibrated probability scale rather than reduction itself. For logistic regression the raw ECE penalty is already small, and for gradient boosting it is negative (reduction *improves* its calibration), so little remains to recover and recalibration leaves the pooled ECE essentially unchanged. However, the deployment-relevant penalties *persist* under recalibration at both depths and for every learner: the pooled AUCR penalty moves only from +0.029 (raw) to +0.028 (Platt) and +0.026 (isotonic), the random forest’s from +0.046 (raw) to +0.038 (Platt) and +0.033 (isotonic), still materially worse in 12 of 12 and 11 of 12 cells respectively, with comparable persistence in the Brier and net-benefit penalties. Notably, isotonic regression—the more flexible recalibrator—does not recover them either, indicating that the failure to recover is not attributable to limited recalibration flexibility. Recalibration thus fixes the part of the problem that is a relabelling of probabilities, but not the part that is a loss of the model’s ability to rank its own errors and to deliver clinical net benefit—the property selective prediction and threshold-based decision-making depend on. For tractability this recalibration-depth control uses a single ranker (mutual information) rather than all three; because the raw AUCR penalty is of comparable magnitude across the three rankers in the main grid, we expect the dissociation to hold for the others, but running the full {none, Platt, isotonic} grid for every ranker is left to future work.

5 Discussion

The audit demonstrates that the discrimination of a reduced model is an insufficient certificate of its safety. Across twelve public clinical datasets, three learners, and three standard rankers, reducing the feature budget was associated, at aggressive cuts, with degraded selective reliability (AUCR, worse in 93 of 108 cells), reduced clinical net benefit (78 of 108), and larger, less informative conformal prediction sets (80 of 108): a loss comparable in extent to, and partly independent of, the loss of discrimination, and present in cells where discrimination remained materially preserved. Probability calibration, by contrast, was mixed across the broader dataset set: aggressive reduction improved and degraded the calibration error with approximately equal frequency, driven by the gradient-boosted learner whose sharp probabilities tightened as redundant features were dropped. The deployment-relevant harms—degradation in the trustworthiness of the model’s confidence and in the net benefit of acting on

Table 7. Raw versus leakage-safe Platt- and isotonic-recalibrated reduction penalty at 25 % of features (mean paired difference vs. full over the twelve datasets, the mutual-information ranker, and three learners; positive ECE/Brier/AURC values indicate degradation, as do negative AUROC/net-benefit values), with the count of materially worse cells. Recalibration recovers the (learner-specific) calibration-error penalty but not the AURC, Brier, or net-benefit loss, and isotonic does no better than Platt. Computed from the released recalibration-control summary.

Outcome	Raw penalty (worse)	Platt penalty (worse)	Isotonic penalty (worse)
AUROC ($\uparrow$)	-0.0327 (32/36)	-0.0338 (33/36)	-0.0311 (30/36)
ECE ($\downarrow$)	-0.0016 (13/36)	+0.0010 (13/36)	+0.0015 (18/36)
Brier ($\downarrow$)	+0.0102 (25/36)	+0.0124 (33/36)	+0.0121 (29/36)
AURC ($\downarrow$)	+0.0294 (32/36)	+0.0275 (32/36)	+0.0259 (30/36)
Net benefit ($\uparrow$)	-0.0098 (27/36)	-0.0117 (29/36)	-0.0109 (29/36)

it—are therefore the most robust signal, and none of them is visible to the AUROC-only evaluation by which feature reduction is usually justified.

Why selective reliability matters here. AURC summarises whether a model’s own confidence can be trusted to triage which predictions to act on. A reduced model with a larger AURC is one whose abstention or referral behaviour is less dependable—a property of direct operational relevance in decision support, and one that neither AUROC nor a single calibration number captures. That AURC degrades early and steeply under reduction (Fig. 2) is the most deployment-relevant property the protocol exposes.

Conditions under which reduction is safe. The extent and direction of degradation are dataset-dependent rather than intrinsic to feature reduction. The cross-dataset meta-analysis (Section 4.7) shows the harm is predicted by the feature count and the intrinsic dimensionality (Spearman $\rho = -0.73$ and $+0.64$): reduction is comparatively safe on wide datasets whose signal is concentrated in a few components—a quarter of the features still spans the predictive subspace—and harmful on compact datasets where each feature carries unique signal. This provides an *a priori* criterion: whether a dataset lies in the safe or the damaging regime can be determined before committing to reduction, rather than only by auditing the reduced model. It also delimits the scope of the claim. We do not assert that feature reduction is universally harmful; rather, aggressive reduction on compact, low-redundancy clinical datasets degrades reliability in ways that discrimination alone does not reveal, and this regime is predictable in advance.

Practical guidance. When feature reduction is being considered, calibration (Brier and ECE), selective reliability (the risk–coverage curve), clinical net benefit (a decision curve), distribution-free set size (conformal efficiency), and subgroup gaps should be reported for each candidate budget alongside discrimination, on the same paired cross-validation splits we use here. Because the harm is heterogeneous but predictable, the decision should be made per dataset and per learner—guided by the feature-count / intrinsic-dimension screen above—rather than by a single global rule. Where the calibration-error component is the primary concern, post-hoc recalibration of the reduced model is an effective and computationally inexpensive mitigation (Section 4.8), and is particularly important for tree ensembles, whose ECE penalty is largely attributable to baseline miscalibration. However, recalibration is not a substitute for retaining the features: it does not restore the selective-reliability (AURC) or Brier loss, so where the model’s confidence will be used to triage or defer, feature reduction itself must be reconsidered rather than corrected post hoc.

Relation to prior work. The individual ingredients—feature rankers, calibration metrics, selective-prediction machinery, subgroup gaps—are all standard, and we deliberately use established forms of each so the protocol inherits their validity. The methodological contribution is their composition into a single pre-registered, leakage-safe certification protocol that treats calibration, selective reliability, and subgroup equity as *dependent variables of the reduction level*, rather than evaluating a final model once on discrimination. It constitutes a reusable evaluation methodology for a recurring clinical-informatics decision—how aggressively to reduce features—assessed on the axes that govern deployment. We do not claim superiority for any classifier; the deliverable is the evaluation methodology and the generalizable lesson it yields.

5.1 Limitations and threats to validity

Construct. ECE is binning-dependent; we fix 15 equal-width bins a priori, treat the binning-free Brier score as the primary calibration outcome (and Brier and ECE agree in direction throughout), and did not run a bin-count sensitivity sweep. AURC uses confidence $|p - 0.5|$, a monotone function of the predicted probability itself; selective reliability and calibration are therefore not fully independent and are expected to co-move, so we do not treat AURC as separate evidence of the calibration

effect—its value is that it degrades while AUROC stays flat, not that it is statistically independent of ECE/Brier. Subgroup gaps use a coarse protected attribute (sex, gender, or a binary age split) and a minimum subgroup size, so they are conservative; following SAGER guidance we note that these attributes are recorded only as binary categories, that sex and gender are not distinguished in the source data, and that a finer sex- and gender-based analysis was not possible with these public datasets. The base learners are reported uncalibrated in the main analysis, which confounds the random forest’s larger ECE penalty with its known baseline miscalibration; the leakage-safe recalibration control (Section 4.8) disambiguates this and shows the forest’s extra ECE penalty is largely a recalibration artifact, while the AURC penalty is not.

Internal. The leakage firewall (in-fold imputation, ranking, scaling, fitting) is enforced by a test that shuffling the labels collapses AUROC to chance. Paired, split-shared comparisons remove between-split variance from the reduction effect.

External. The twelve public datasets cover several clinical contexts but are mostly modest in size ($n = 155\text{--}6,000$, with only the readmission cohort large) and low-to-moderate dimensional ($p = 3\text{--}44$), and are not prospective; absolute effect sizes may differ on larger or temporally split cohorts. In particular, the high-dimensional regime ($p \gg 44$, e.g. genomics or wide EHR feature sets) where aggressive reduction is most common is out of scope here, and the claims are scoped to low/moderate-dimensional tabular clinical data. The protected attributes available limit the fairness analysis to the recorded groups, and the Diabetes-130 readmission cohort—the study’s most consequential fairness context—is deterministically subsampled to 6,000 encounters for tractability, which reduces statistical power for the smallest race subgroups; its fairness numbers should be read as indicative rather than definitive.

Statistical. Wilcoxon plus Holm controls family-wise error within each (dataset, method, learner, outcome) family across the feature budgets; the aggregate cell counts we report are descriptive summaries across those families, not a single study-wide family-wise test. The cross-dataset meta-analysis ($n = 12$) is exploratory; only the feature-count predictor survives multiplicity correction. Bootstrap CIs quantify effect-size uncertainty, and we report materiality (significance *and* a CI excluding zero), not significance alone.

6 Conclusion

In a pre-registered, leakage-controlled audit on twelve public clinical datasets, aggressive feature reduction degraded selective reliability, clinical net benefit, and conformal informativeness at least as much as discrimination—dose-dependently, most severely for random forests, and in a set of settings while discrimination remained materially preserved—while probability calibration was mixed across datasets and the subgroup gaps moved in no systematic direction. Importantly, the harm was strongly conditional on the dataset yet *predictable*: a cross-dataset meta-analysis shows it is governed by the feature count and intrinsic dimensionality, so reduction is comparatively safe on wide, low-intrinsic-dimension datasets and harmful on compact ones. The practical conclusion is that calibration, selective reliability, net benefit, and subgroup gaps should be audited explicitly, per dataset and per learner, before a reduced feature set is adopted, as the discrimination of the reduced model does not reveal them, and that the feature-count and intrinsic-dimension screen can identify the damaging regime in advance. Post-hoc recalibration repairs the calibration-error component but not the selective-reliability or net-benefit loss, and is therefore a limited mitigation. Future work should include prospective and temporally split validation. All code, the calibration file, the pre-registration, and the per-configuration summary are released so that every number regenerates from a recorded seed.

Ethics statement

This study used only publicly available, fully de-identified secondary datasets and involved no intervention or interaction with human participants and no access to identifiable private information. It therefore does not constitute human-subjects research and was exempt from institutional review board approval.

Data availability

All twelve datasets are public and were retrieved in June 2026. Ten are from the UCI Machine Learning Repository, cited by dataset identifier: Cleveland heart disease (45), Statlog heart (145), SPECTF heart (96), Hepatitis (46), Indian Liver Patient (225), Mammographic Mass (161), Haberman survival (43), Breast Cancer Wisconsin original (15), and the Diabetes 130-US hospitals readmission cohort (296, <https://archive.ics.uci.edu/dataset/296>); the Heart-Failure clinical records dataset is UCI 519. The Pima Indians diabetes data (originally Smith et al., 1988) and the Wisconsin Diagnostic Breast Cancer dataset (originally Street et al., 1993; bundled with the scikit-learn library) are likewise public. The Diabetes-130 cohort is deterministically stratified-subsampled to $n = 6,000$ (seed recorded) for tractability. Per-dataset provenance (source, URL, retrieval date, preprocessing) is documented in the project repository. Raw files are not redistributed; the released code reproduces the exact preprocessing.

Code availability

All code, the calibration file, the pre-registration, and the per-configuration result summaries are available in the project repository at <https://github.com/helghareeb/ns-clinical-fs>, with a documented reproduction entry point; every number in this paper regenerates from committed code, the calibration file, and the recorded seed. The repository is currently private and is available to the editors and reviewers for inspection on request during peer review; it will be made publicly accessible upon acceptance.

Sex and gender

Sex (Cleveland, Statlog and Heart-Failure heart, Hepatitis), gender (ILPD), an age stratum (Pima, Haberman, Mammographic), and race (the Diabetes-130 readmission cohort, six groups) are used in this study *only* as protected attributes for the subgroup-equity analysis and are never model features; the WDBC, original breast-cancer, and SPECTF datasets carry no protected attribute. We report between-subgroup calibration, discrimination, net-benefit, and conformal-coverage gaps as a function of feature reduction. Following SAGER guidance, we note as a limitation that most attributes are recorded as coarse binary categories, that sex and gender are not distinguished in the source data, and that a finer sex- and gender-based analysis was not possible with these datasets.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Competing interests

The author declares no competing interests.

Author contributions

H.A.E. is the sole author. H.A.E. conceived and designed the study, implemented the software, performed the experiments, analysed the data, prepared the figures, and wrote and reviewed the manuscript.

Use of large language models

During the preparation of this work the author used an AI coding assistant (Claude Code) in order to support software-engineering and manuscript-preparation tasks. After using this tool the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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